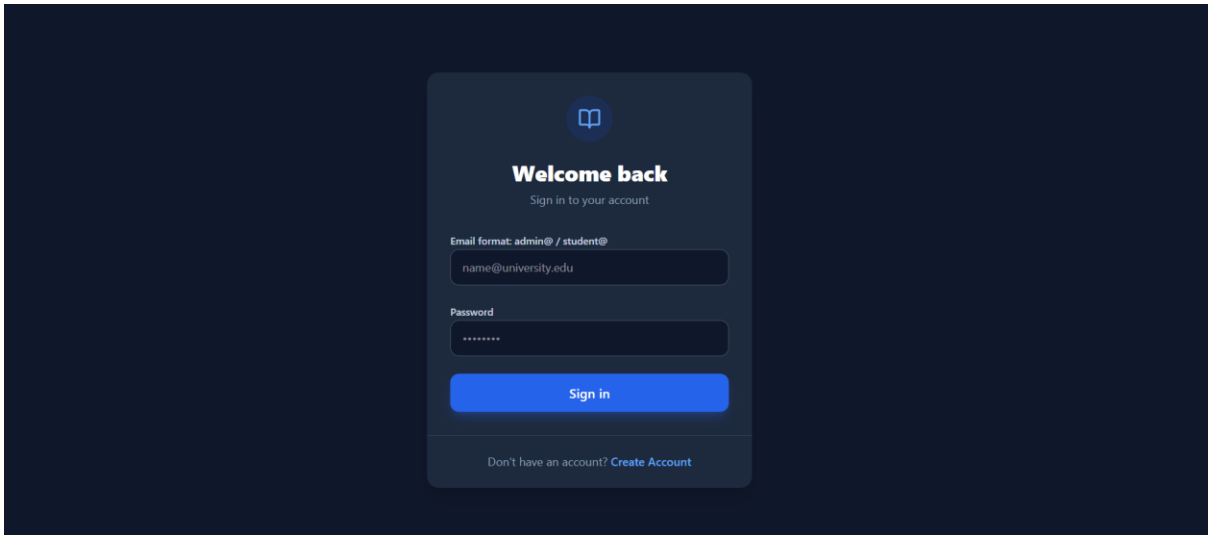
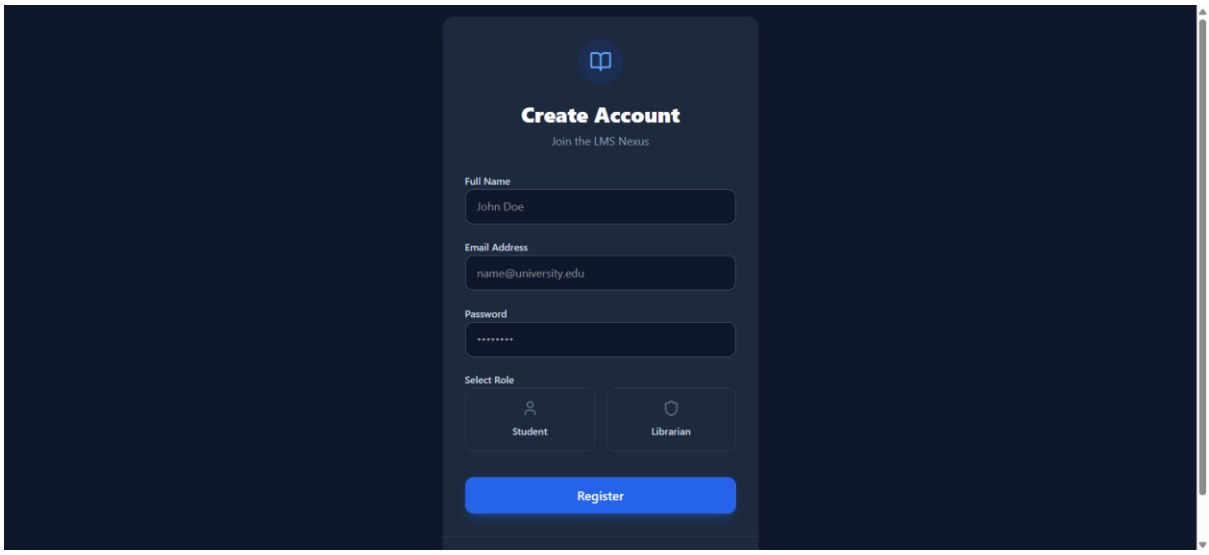
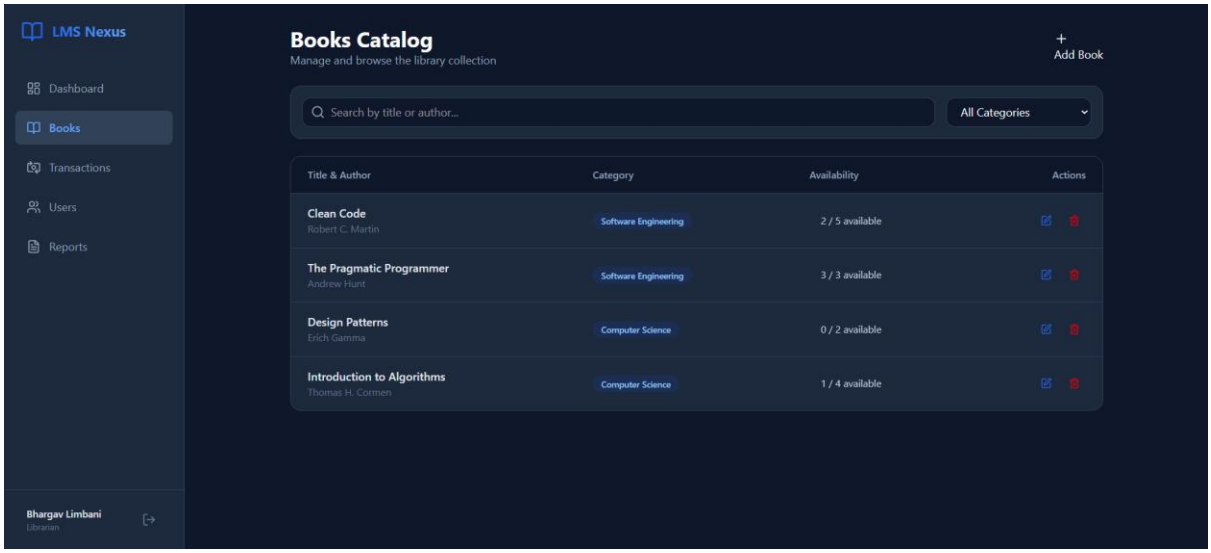
Bhargav Limbani 92301733029

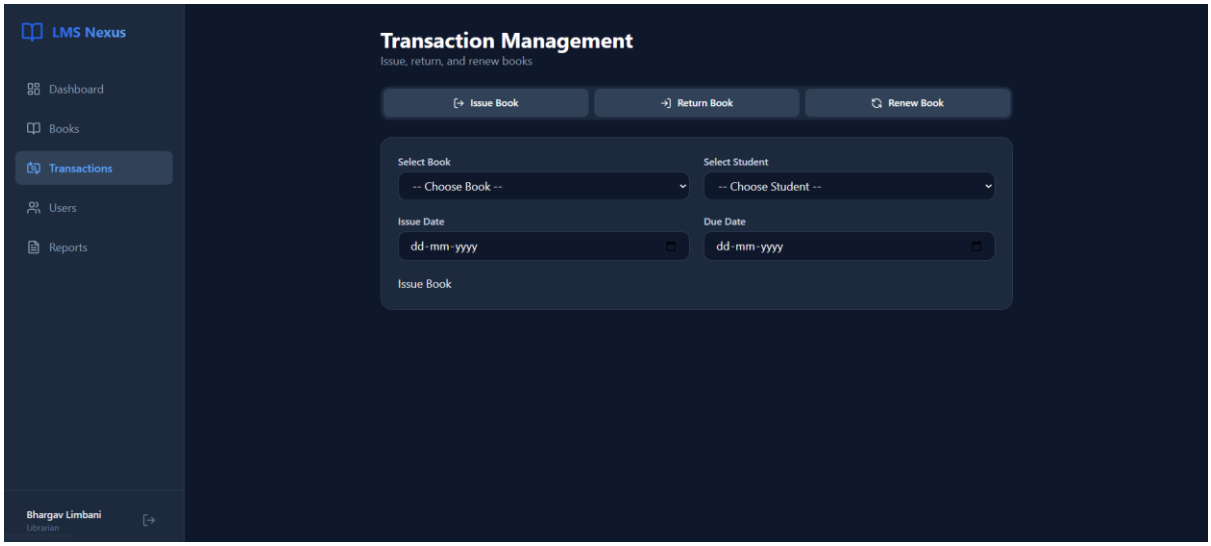
Take Home Test-1

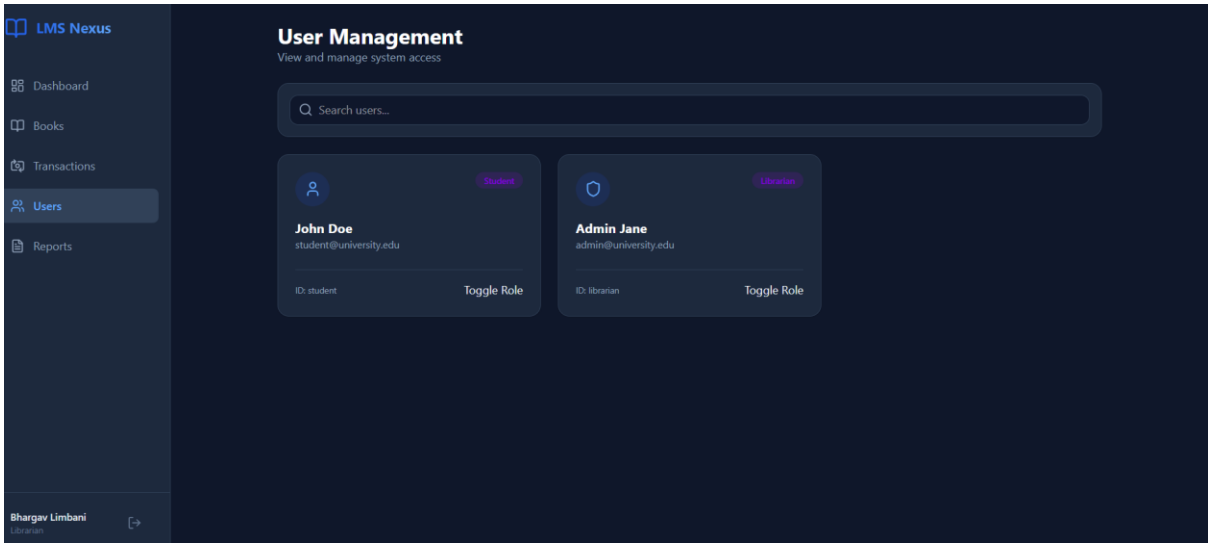
Sr. No.	Section	Task Description	CO Mapped
1	Front End Design (React)	Initialize the React project using Vite, install dependencies (react-router-dom, axios, react-hook-form, Tailwind CSS, recharts), and create a clean folder structure (features/, shared/components/, contexts/, services/, utils/) following clean architecture principles for separation of concerns in UI layer.	3, 4
2	Front End Design (React)	Develop authentication UI components: Login and Register pages with form validation (react-hook-form), role selection (student/librarian), error handling, and Context API setup for initial auth state (token/role stored in localStorage).	4
3	Front End Design (React)	Create component-based Book Management UI: BookList (search/filter/pagination using React hooks), AddBook/EditBook modal forms (protected via role context), following clean component composition and reusable UI elements.	4
4	Front End Design (React)	Build User Management components: StudentDashboard (display personal borrowed books/transactions), LibrarianUserList (view/edit users with role-based rendering), using React Router for protected routes.	3, 4
5	Front End Design (React)	Design Transaction UI components: IssueBookForm, RenewBookForm, ReturnBookForm with date pickers for due dates, availability preview, and fine preview; implement clean custom hooks for form logic.	3, 4
6	Front End Design (React)	Implement Reports UI: ReportDashboard with tabs for various reports (overdue books table, issued books chart via recharts, transaction history); use React state and reusable table components (mock data for now).	3, 4


Welcome back
Sign in to your account
Email format: admin@ / student@
name@university.edu
Password
Sign in
Don't have an account? Create Account

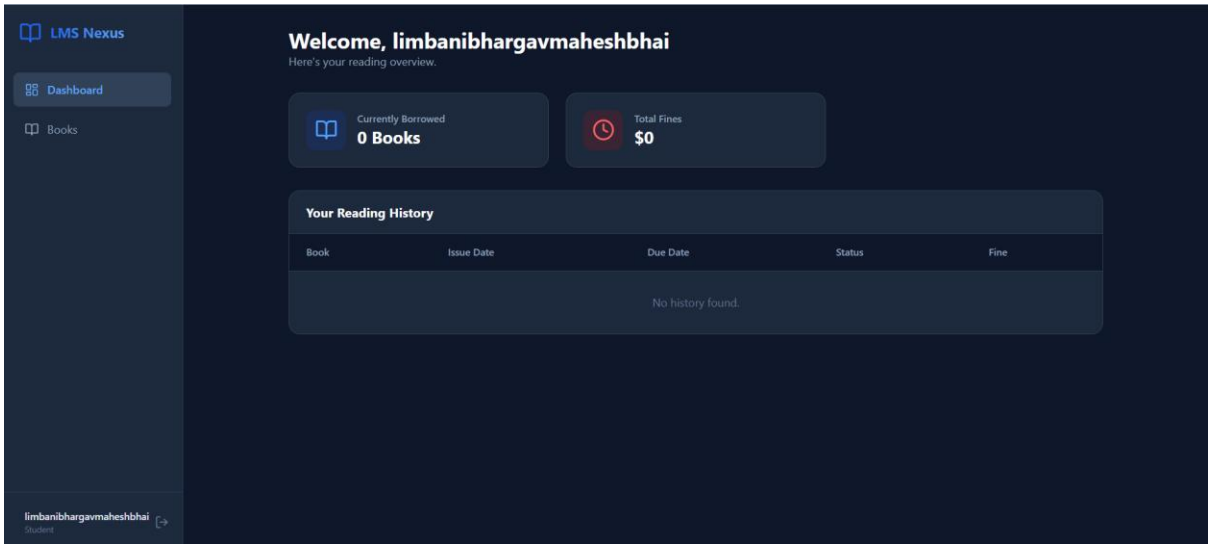

Create Account
Join the LMS Nexus
Full Name
John Doe
Email Address
name@university.edu
Password
Select Role
Student
Librarian
Register


LMS Nexus
Dashboard
Books
Transactions
Users
Reports
Bhargav Limbani
Librarian
Books Catalog
Manage and browse the library collection
Add Book
Search by title or author...
All Categories
Title & Author
Category
Availability
Actions
Clean Code
Robert C. Martin
Software Engineering
2 / 5 available
The Pragmatic Programmer
Andrew Hunt
Software Engineering
3 / 3 available
Design Patterns
Erich Gamma
Computer Science
0 / 2 available
Introduction to Algorithms
Thomas H. Cormen
Computer Science
1 / 4 available


LMS Nexus
Dashboard
Books
Transactions
Users
Reports
Bhargav Limbani
Librarian
Transaction Management
Issue, return, and renew books
Issue Book
Return Book
Renew Book
Select Book
-- Choose Book --
Select Student
-- Choose Student --
Issue Date
dd-mm-yyyy
Due Date
dd-mm-yyyy
Issue Book


LMS Nexus
Dashboard
Books
Transactions
Users
Reports
Bhargav Limbani
Librarian
User Management
View and manage system access
Search users...
Student
John Doe
student@university.edu
ID: student
Toggle Role
Librarian
Admin Jane
admin@university.edu
ID: librarian
Toggle Role

LMS Nexus
Dashboard
Books
Transactions
Users
Reports
Bhargav Limbani
Librarian
Reports Overview
Library activity and analytics
Issue & Return Trends
1
0.75
0.5
0.25
0
2023-09
2023-10
Overdue Books (1)
Book Title
User ID
Due Date
Days Overdue
Clean Code
student
2023-10-15
916 days


LMS Nexus
Dashboard
Books
limbanibhargavmaheshbhai
Student
Welcome, limbanibhargavmaheshbhai
Here's your reading overview.
Currently Borrowed
0 Books
Total Fines
$0
Your Reading History
Book
Issue Date
Due Date
Status
Fine
No history found.